

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing customer shopping behavior using transactional data collected from 3,900 purchases across different product categories. The purpose of the analysis is to identify trends in customer spending, understand buying habits, explore product preferences, and evaluate subscription-related behavior. The insights gathered can help support better business and marketing decisions.

2. Dataset Summary

The dataset contains 3,900 records and 18 variables related to customer demographics, purchasing activity, and shopping behavior.

Key Features Included

- Customer information such as age, gender, location, and subscription status
- Purchase-related details including item purchased, category, purchase amount, season, size, and color
- Shopping behavior indicators such as discounts applied, promo code usage, previous purchases, purchase frequency, review ratings, and shipping type

Data Quality

A small number of missing values were identified in the Review Rating column, with 37 records containing incomplete entries.

3. Exploratory Data Analysis Using Python

The analysis process started with data preparation and initial exploration using Python.

Steps Performed

- Imported the dataset using the pandas library
- Looked at data `df.head()`
- Reviewed the dataset structure using `df.info()`

- Generated descriptive statistics with `df.describe()` to better understand the numerical variables
- Checked the dataset for missing values and overall data consistency before proceeding with further analysis

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ver
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	C
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cr
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pa
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pa

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe(include = 'all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

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Data Cleaning and Preparation

- Missing values were identified and handled appropriately. The missing entries in the `review_rating` column were filled using the median rating within each product category to maintain consistency in the data.
- Column names were standardized by converting them into `snake_case` format, which improved readability and made documentation and querying easier.

Feature Engineering

Additional features were created to support deeper analysis:

- An `age_group` column was introduced by grouping customer ages into ranges.
- A `purchase_frequency_days` feature was created to better understand customer purchasing patterns over time.

Data Validation

A consistency check was conducted to determine whether `discount_applied` and `promo_code_used` provided overlapping information. Since the two variables were found to be largely redundant, the `promo_code_used` column was removed to simplify the dataset.

Database Integration

After cleaning and preprocessing, the dataset was connected to a MySQL database through Python. The cleaned DataFrame was then loaded into the database to support further analysis using SQL.

4. Data Analysis Using SQL (Business Transactions)

Structured analysis was performed in MySQL to answer important business-related questions and identify customer trends.

1. Revenue by Gender

A comparison was conducted between male and female customers to evaluate which group generated higher overall revenue and contributed more to total sales.

	gender	revenue
►	Male	157890
	Female	75191

2. High-Spending Discount Users

The analysis identified customers who made use of discounts while still spending more than the overall average purchase amount. This helped highlight a group of high-value customers who remained strong spenders even when promotional offers were applied.

Result Grid		Filter Rows:
	customer_id	purchase_amount
►	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62

3. Top 5 Products by Rating

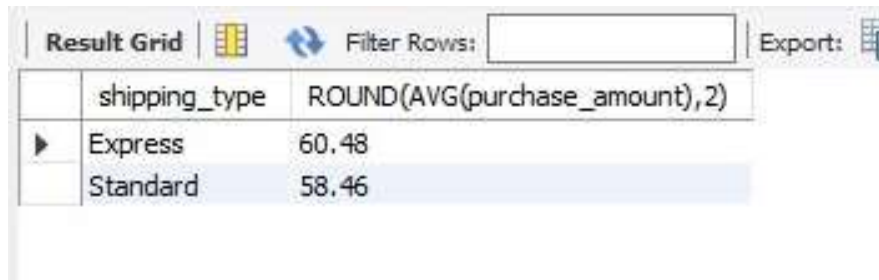
The analysis identified the top five products with the highest average customer review ratings. This provided insight into the products that customers were most satisfied with and helped highlight items that performed well in terms of customer experience and product quality.



item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. Shipping Type Comparison

A comparison was conducted between Standard and Express shipping methods to evaluate differences in average purchase amounts. The analysis helped determine whether customers choosing faster delivery options tended to spend more than those selecting standard shipping.



shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. Subscribers vs. Non-Subscribers

The analysis compared subscribers and non-subscribers based on their average spending and overall revenue contribution. This helped assess the value of subscription-based customers and understand how subscription status influenced purchasing behavior.

Result Grid				
	subscription_status	total_customers	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

6. Discount-Dependent Products

The analysis identified the five products with the highest proportion of purchases made using discounts. This helped reveal which products were most dependent on promotional offers to drive sales and customer purchases.

Result Grid		
	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. Customer Segmentation

Customers were grouped into different segments based on their purchase history and shopping activity. The segments included New, Returning, and Loyal customers, allowing for a clearer understanding of customer retention patterns and purchasing behavior across different groups.

Result Grid		
	customer_segment	Number of Customers
▶	Loyal	3116
	Returning	701
	New	83

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8. Top 3 Products per Category

The analysis identified the three most purchased products within each product category. This provided insight into customer preferences and highlighted the best-performing items across different categories.

Result Grid				
		Filter Rows:	Export: 	
	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

9. Repeat Buyers and Subscriptions

The relationship between repeat purchases and subscription status was examined by analyzing customers with more than five purchases. The goal was to determine whether frequent buyers were more likely to subscribe, providing insight into customer loyalty and subscription engagement.

Result Grid		
		Filter Rows: 
	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

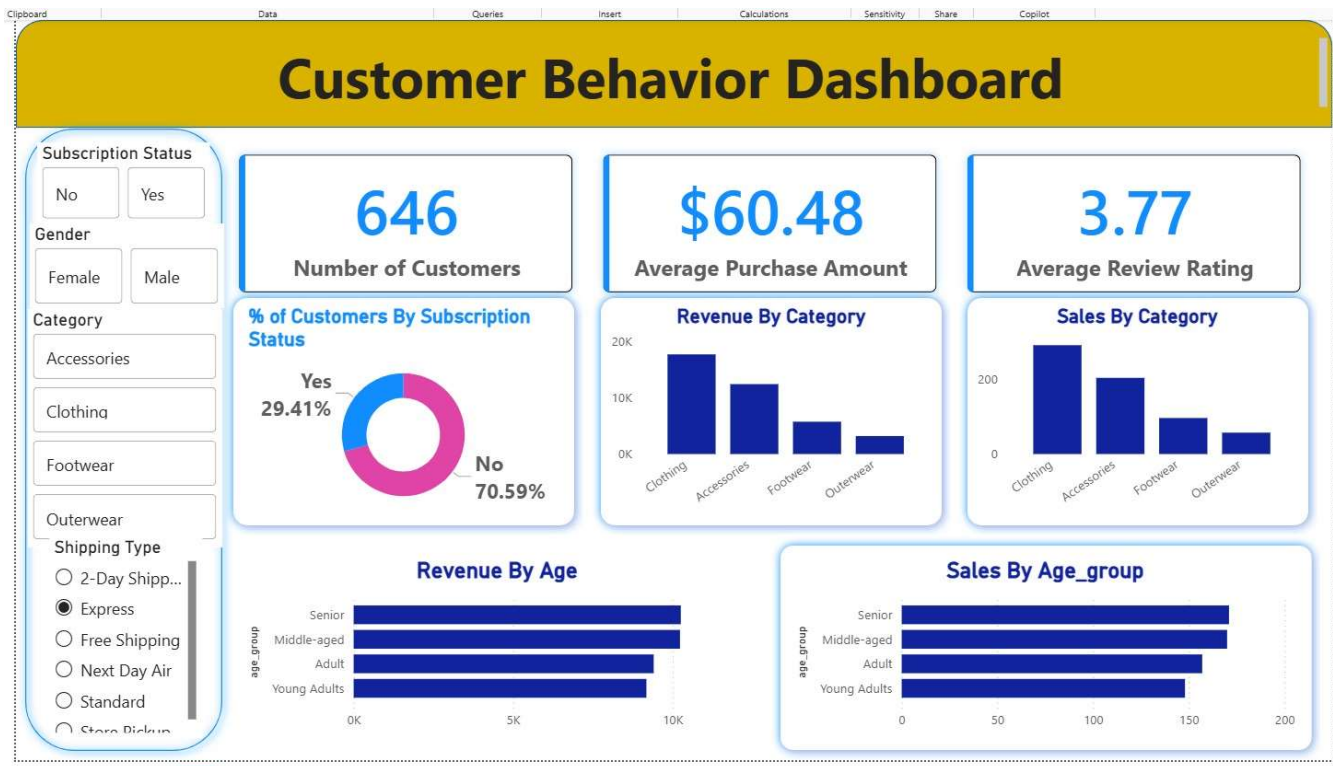
10. Revenue by Age Group

The analysis calculated the total revenue generated by each age group to better understand which customer segments contributed the most to overall sales. This helped identify the age categories with the highest purchasing power and spending activity.

Result Grid			Filter Rows:
	age_group	total_revenue	
▶	Young Adults	62143	
	Middle-aged	59197	
	Adult	55978	
	Senior	55763	

5. Dashboard Development in Power BI

To present the findings in a clear and interactive way, a dashboard was developed in Power BI. The dashboard provided visual insights into customer behavior, spending patterns, product performance, subscription trends, and overall business performance, making it easier to explore and interpret the data.



6. Business Recommendations

- **Increase Subscription Engagement**

Encourage more customers to subscribe by offering exclusive benefits such as special discounts, early access to products, or loyalty rewards.

- **Strengthen Customer Loyalty Programs**

Introduce or improve loyalty programs that reward repeat purchases and encourage customers to progress into the “Loyal” customer segment.

- **Evaluate Discount Strategies**

Review the current discount approach to ensure promotions continue driving sales without significantly affecting profit margins.

- **Improve Product Positioning**

Feature top-rated and best-selling products more prominently in marketing campaigns and promotional activities to increase visibility and sales performance.

- **Enhance Targeted Marketing Efforts**

Focus marketing campaigns on high-revenue customer age groups and customers who frequently choose express shipping, as these groups showed stronger spending behavior.